

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460782.68 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1802 +/- 0.0083
Transit depth (Rp/Rs)^2	3.25 +/- 0.3 [%]
Orbital Inclination (inc)	90.0 +/- 0.54
Airmass coefficient 1 (a1)	1.424 +/- 0.017
Airmass coefficient 2 (a2)	-0.1716 +/- 0.0055
Scatter in the residuals of the lightcurve fit is	1.2 %
Best Comparison Star	None
Optimal Aperture	5.09
Optimal Annulus	8.55
Transit Duration (day)	0.09721 +/- 0.00089

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1802 ± 0.0083 vs 0.1284 (deviation 5.73σ)
Duration fit vs published	0.09721 d vs 0.0775 d (deviation 20.34σ)
Mid-time O–C	-3.8 min
Depth SNR	19.9
Residual scatter	1.2 %

Light curve

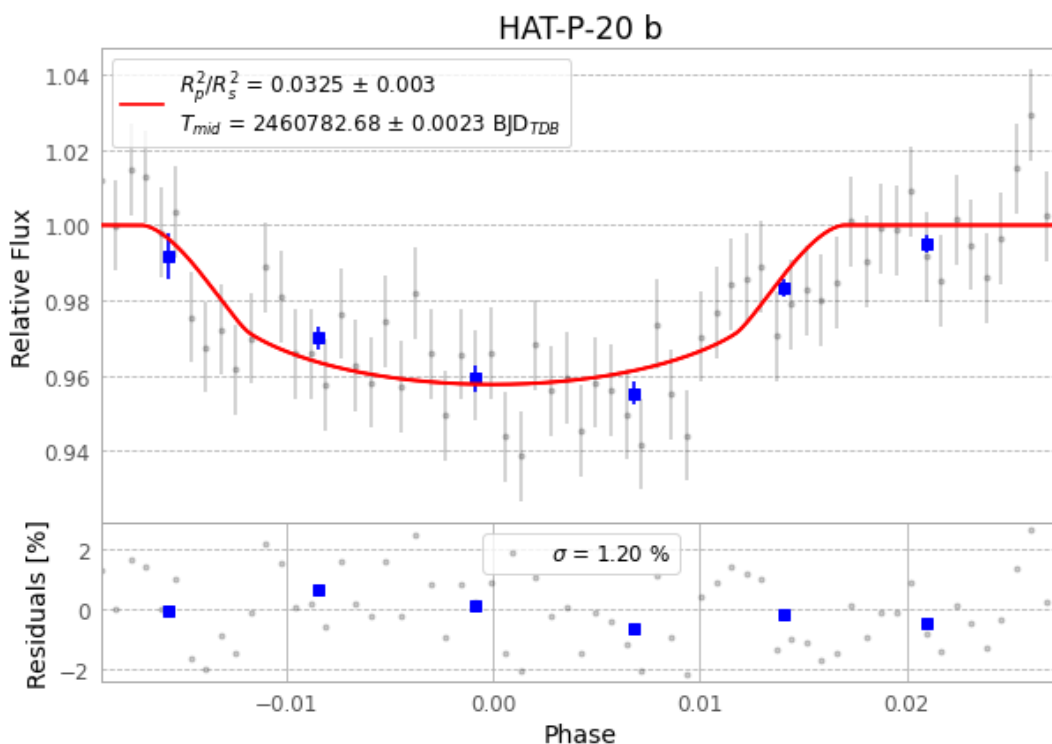


Figure 1 — FinalLightCurve_HAT-P-20 b_2025-04-16.png (SHA-256 76baad3d10226f0b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [296, 259], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 96e3564acee44cc4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

65 FITS frames (43 MB), HATP-20250417030015.FITS → HATP-20250417061215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-20-250417.log (24efe6a34b36...), FinalLightCurve_HAT-P-20 b_2025-04-16.png (76baad3d1022...), FinalLightCurve_HAT-P-20 b_2025-04-16.csv (69580a4fd046...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 20:16Z.